Literature Search and Selection Record

Retrospective methodological record supporting Chapter 2, Section 2.2. Search date: **11 September 2026**.

1. Scope and search sources

A focused thematic literature search identified research on large language model (LLM) reliability, retrieval-augmented generation (RAG), educational RAG, AI assistance for programming students, and related research methodology. Searches were conducted through Google Scholar and Semantic Scholar. Transparency and evidence provenance were considered across these themes. This was a targeted critical review, not a formal systematic review.

The principal search expressions documented for the project are listed below; additional combinations were considered during source discovery.

Search theme	Recorded representative search expressions	Purpose
LLM reliability	“LLM hallucination”; “hallucination taxonomy”	Factual reliability, source faithfulness and error types
RAG architecture	“retrieval-augmented generation”; “RAG failure modes”	Grounding mechanisms, retrieval limitations and attribution
Educational RAG	“RAG chatbot education”; “educational chatbot”	Course-based retrieval and educational evidence
Programming education	“ChatGPT programming education”; “student trust AI programming”	Curricular fit, programming help and learner perceptions
Research methodology	“Design Science Research”; “DSRM information system”	Artefact-centred research methodology

2. Screening and selection

A shortlist of **26 publications** was assessed for research-question relevance, publication credibility, fit with the defined scope, and a preference for works published from 2020 onwards. The recorded decisions resulted in **17 scholarly publications** retained for direct discussion in Chapter 2 and **nine non-retained candidates**.

26	9	17
Initially shortlisted	Not retained	Retained in Chapter 2

Peer-reviewed work was prioritised. Sources outside the question-answering scope, items without verified bibliographic identity, and publications whose evidence was less directly relevant than selected alternatives were not retained. Publication year was a preference rather than an absolute exclusion rule.

Foundational Design Science Research references, including Hevner et al. (2004) and Peffers et al. (2007), support Chapter 3 and are not included in the 17-publication Chapter 2 set.

3. Recorded non-retention decisions

The following table records the exclusion or non-selection reasons supplied for nine sources considered during literature selection.

Source	Decision basis	Recorded rationale
Feldman et al. (2024)	Publication status	Recorded as arXiv-only, without an identified peer-reviewed publication at the time of selection.
Sharma et al. (2024)	Publication status	Recorded as arXiv-only, without an identified peer-reviewed publication at the time of selection.
Becker et al. (2023)	Relevance to research scope	Broader discussion of generative AI in introductory programming and curricular practice, rather than empirical assessment of course-grounded RAG, citation faithfulness or external fallback; more directly aligned studies were retained.
Karpukhin et al. (2020)	Technical scope	Dense Passage Retrieval internals were beyond the depth needed; Lewis et al. (2020) provided the relevant RAG foundation.
Wong et al. (2023)	Topic scope	Review centred on code generation and Big Code; code generation was outside the project scope. Not part of the approved reading list.
Faraon et al. (2025)	Bibliographic verification	A matching publication could not be verified from the recorded citation details, so the claim was not attributed to this source.
Zawacki-Richter et al. (2019)	Period / relevance	Outside the preferred 2020-onward period and not essential to the selected analytical coverage; the period was not treated as an absolute cutoff.
Kasneci et al. (2023)	Selection overlap	Peer-reviewed and relevant, but not selected for the approved set; Yan et al. (2024) and Bond et al. (2024) covered the chosen educational-AI concerns.
Sheese et al. (2024)	Selection overlap	Identifiable ACE 2024 publication, but not approved for the final set; selected literature was preferred for the review’s defined themes.

4. Publications retained for Chapter 2

Seventeen scholarly publications are grouped by principal analytical role. Full bibliographic details appear in the dissertation reference list.

Analytical strand	Publication	Role in the literature review	§
LLM reliability	Bender et al. (2021)	Fluency and modelling limitations	2.4
	Ji et al. (2023)	Hallucination and source faithfulness	2.3–2.4
	Sun et al. (2024)	Generated-error taxonomy and limitations	2.4
RAG mechanisms	Lewis et al. (2020)	Foundational retrieval-augmented generation	2.5
	Barnett et al. (2024)	Seven failure points; absent versus missed evidence	2.5
	Klesel and Wittmann (2025)	Modular architecture and limitations	2.5
	Byun et al. (2024)	Retrieved-context faithfulness	2.5
	Gao et al. (2023)	Claim-level citation and attribution quality	2.5
Educational RAG	Lang and Gürpinar (2025)	Course coverage and student perceptions	2.6
	Thüs et al. (2024)	Engagement and measured learning outcomes	2.6
	Wijaya and Purwarianti (2024)	Programming tutor and retrieval evaluation	2.6
Programming AI	Kazemitabaar et al. (2024)	CodeAid correctness and pedagogical fit	2.7
	Penney et al. (2025)	Novice reliability and verification concerns	2.7
	Haindl and Weinberger (2024)	Programming-course student experiences	2.7
	Najimi et al. (2026)	Java feedback and teaching constraints	2.7
AI transparency	Yan et al. (2024)	Stakeholder transparency in educational AI	2.8
	Bond et al. (2024)	Ethics and methodological rigour	2.8

5. Contribution to the research gap

The retained publications were synthesised to distinguish factual correctness, support from the fixed course corpus, citation faithfulness, source transparency and student perceptions. These distinctions informed Chapter 2’s research gap (§2.10), CurriculumAI’s two-tier design and its separate technical and student evaluations.

Reproducibility record. The dated search details, selection criteria, exclusion reasons and retained-publication mapping are documented here to support transparency and reproducibility of the literature-selection process.